

Predicting Type II Diabetes Risk Using Explainable Machine Learning

A Healthcare AI Approach for Early Risk Identification using BRFSS 2023 Data



Presented by: Apex

Why Diabetes? The Global Challenge



 589M

Adults living with diabetes worldwide

 252M

Adults remain undiagnosed globally

 813,300

Adults living with diabetes in Kenya

 3.1%

Adult diabetes prevalence in Kenya



Source: IDF Diabetes Atlas, 2021

The Clinical Problem: Late Diagnosis Impact



Undiagnosed Patient

Millions unaware of their risk



Delayed Detection

Symptoms appear years too late

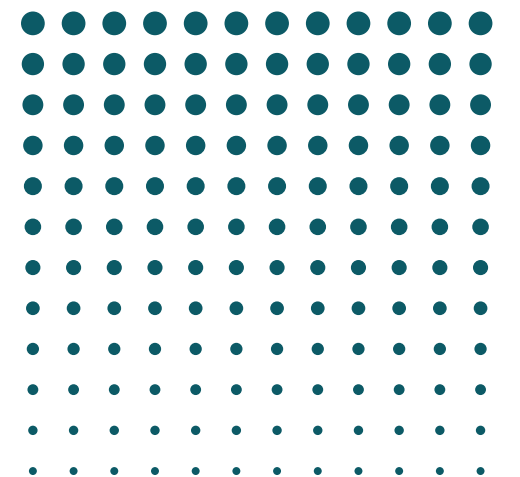
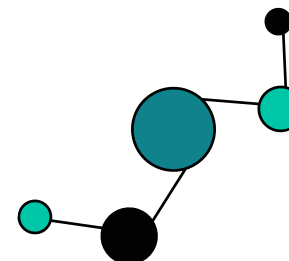


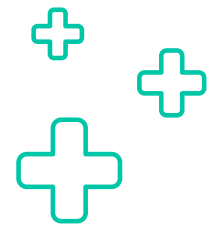
Complications & Hospitalization

Neuropathy, retinopathy, organ damage



Can AI predict risk earlier?





Project Objectives



Explore demographic, behavioral, and clinical factors associated with diabetes.

Train and compare Machine Learning Models.

Identify the most influential predictors and translate them into actionable recommendations.

Implement SHAP and LIME for model explainability

develop an interactive dashboard for disease risk prediction and visualization



Primary Stakeholders:

Clinicians and healthcare providers
Heath promotion programs
Individuals at risk of T2DM



Secondary Stakeholders:

Research and Academic institutions
Health Insurance Organizations
Policy makers
NGOs and Community Health Organizations



Dataset Overview: CDC BRFSS 2023



CDC Behavioral Risk Factor Surveillance System: 433,323 participants surveyed → 350 variables collected → 429,086 final records after cleaning. The largest telephone health survey system in the world.



Data Source

Data Info

Final Data

Features Selected

Target variable

CDC BRFSS 2023 Behavioral Risk Factor Surveillance Survey.

433,323 participants
350 variables collected

4429,086 final records retained after preprocessing.

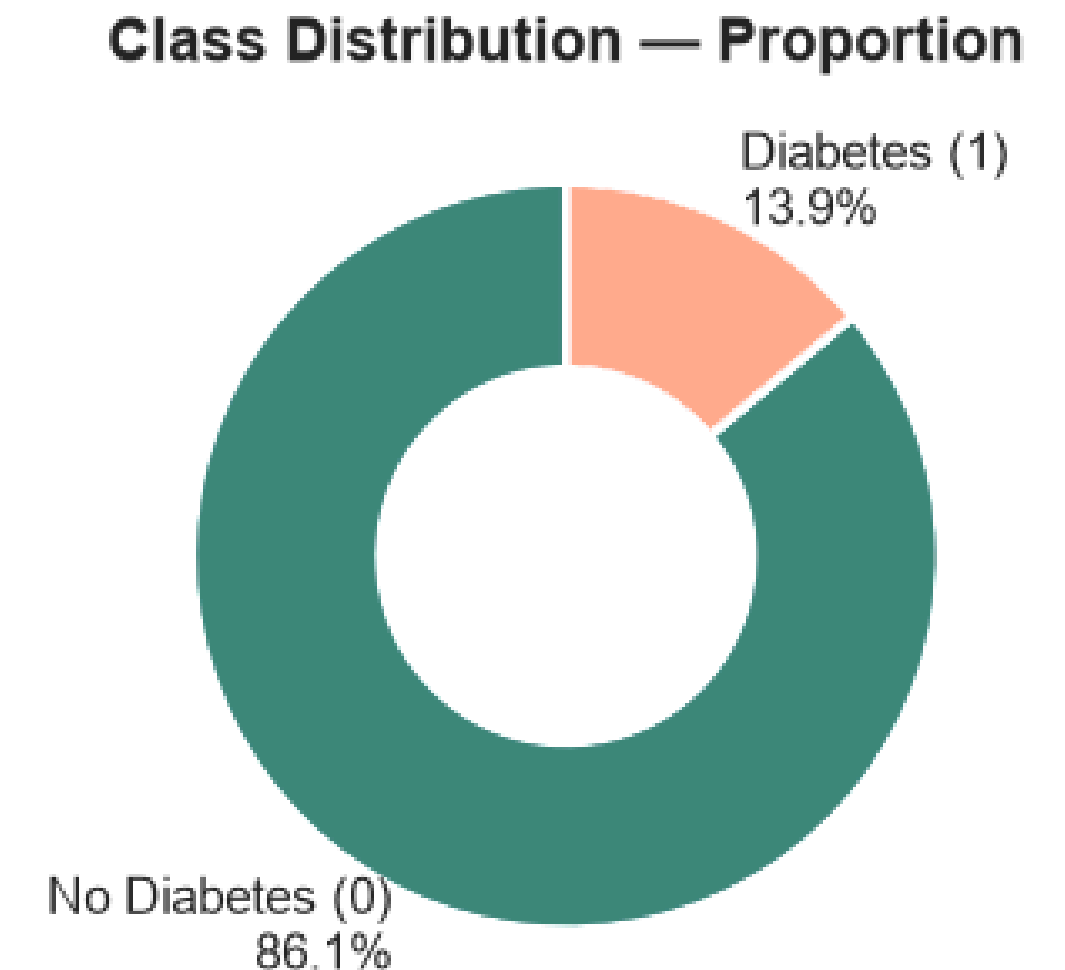
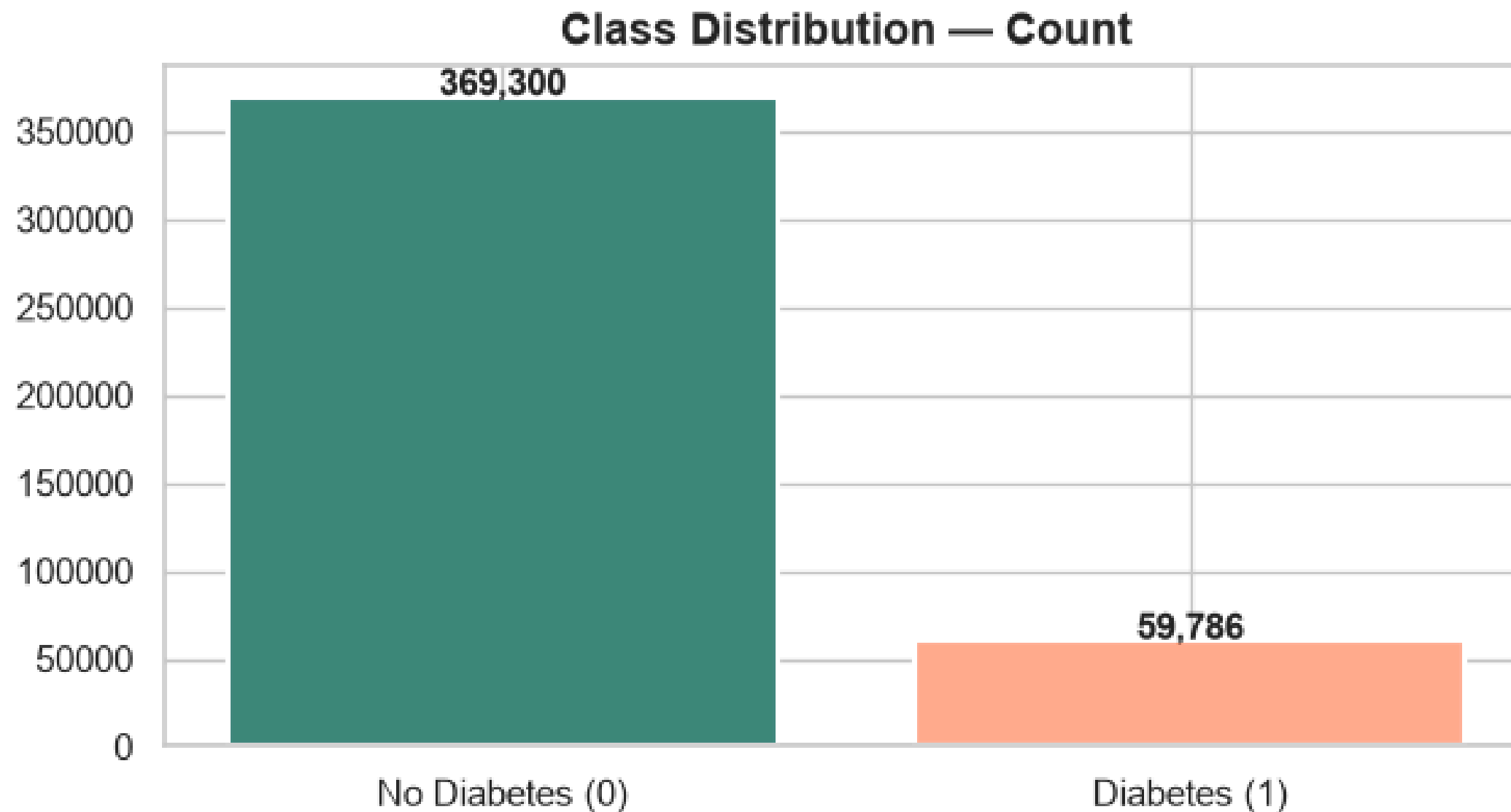
15 key features selected

Type II Diabetes diagnosis (binary).

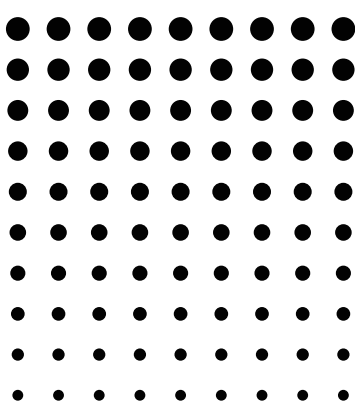
Data Source CDC / BRFSS 2023

Target Variable Distribution

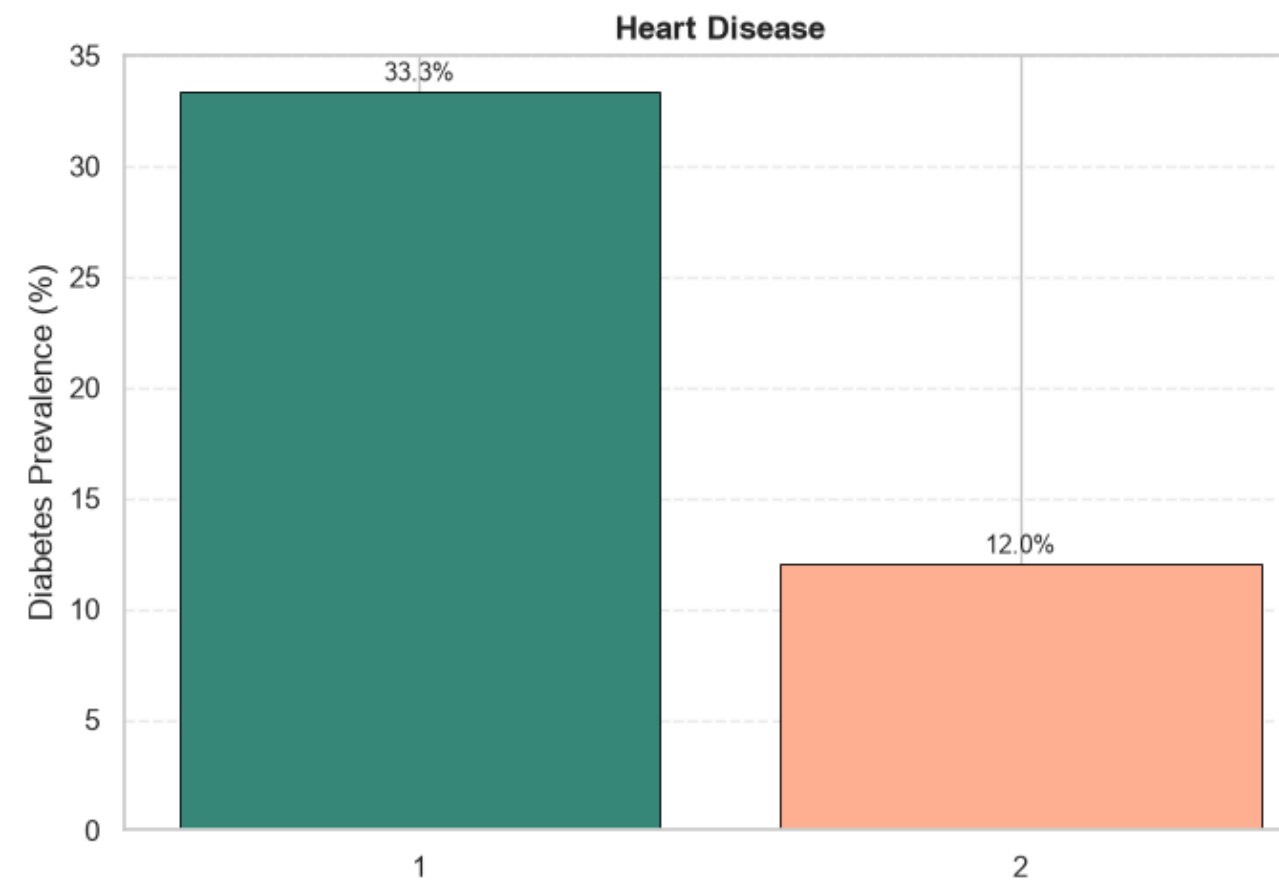
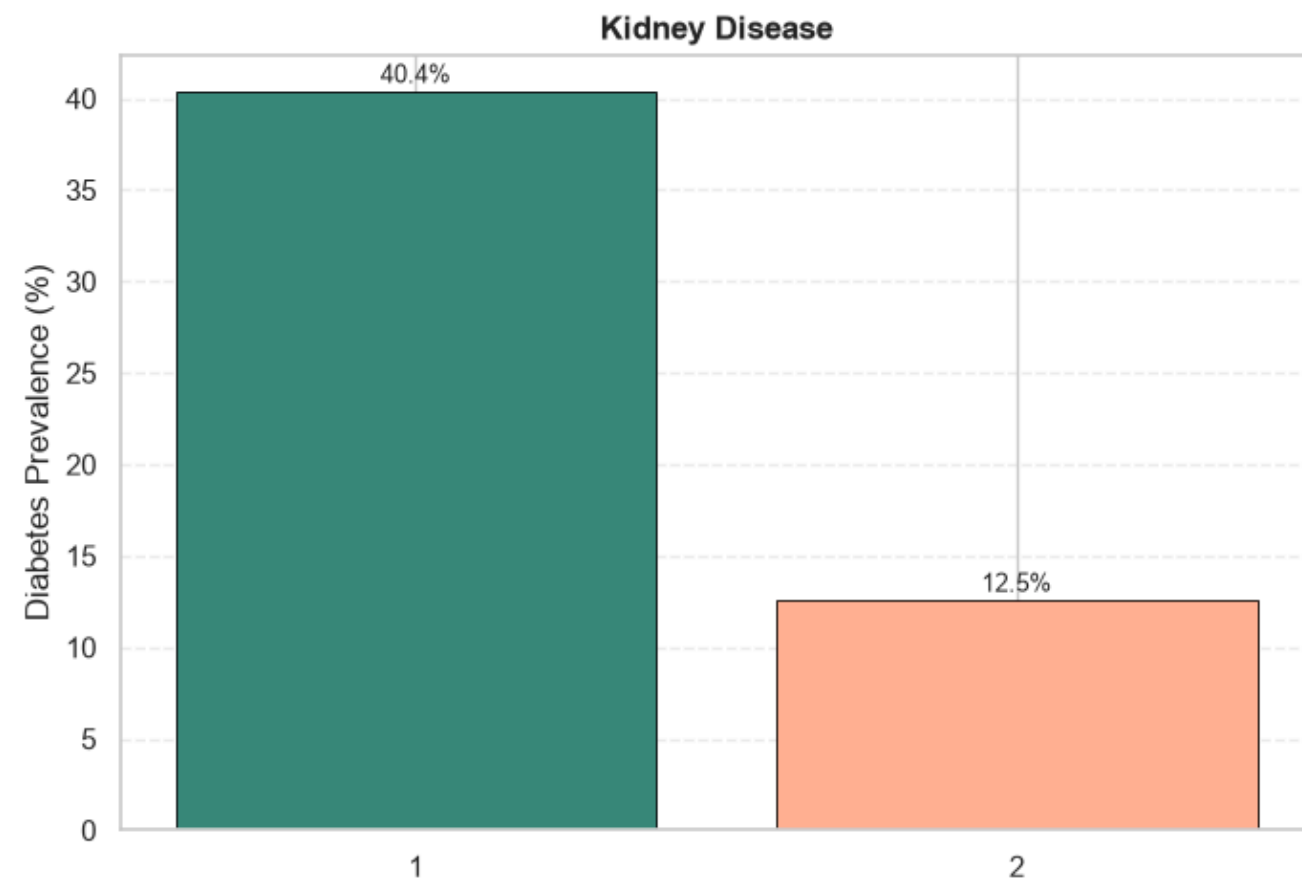
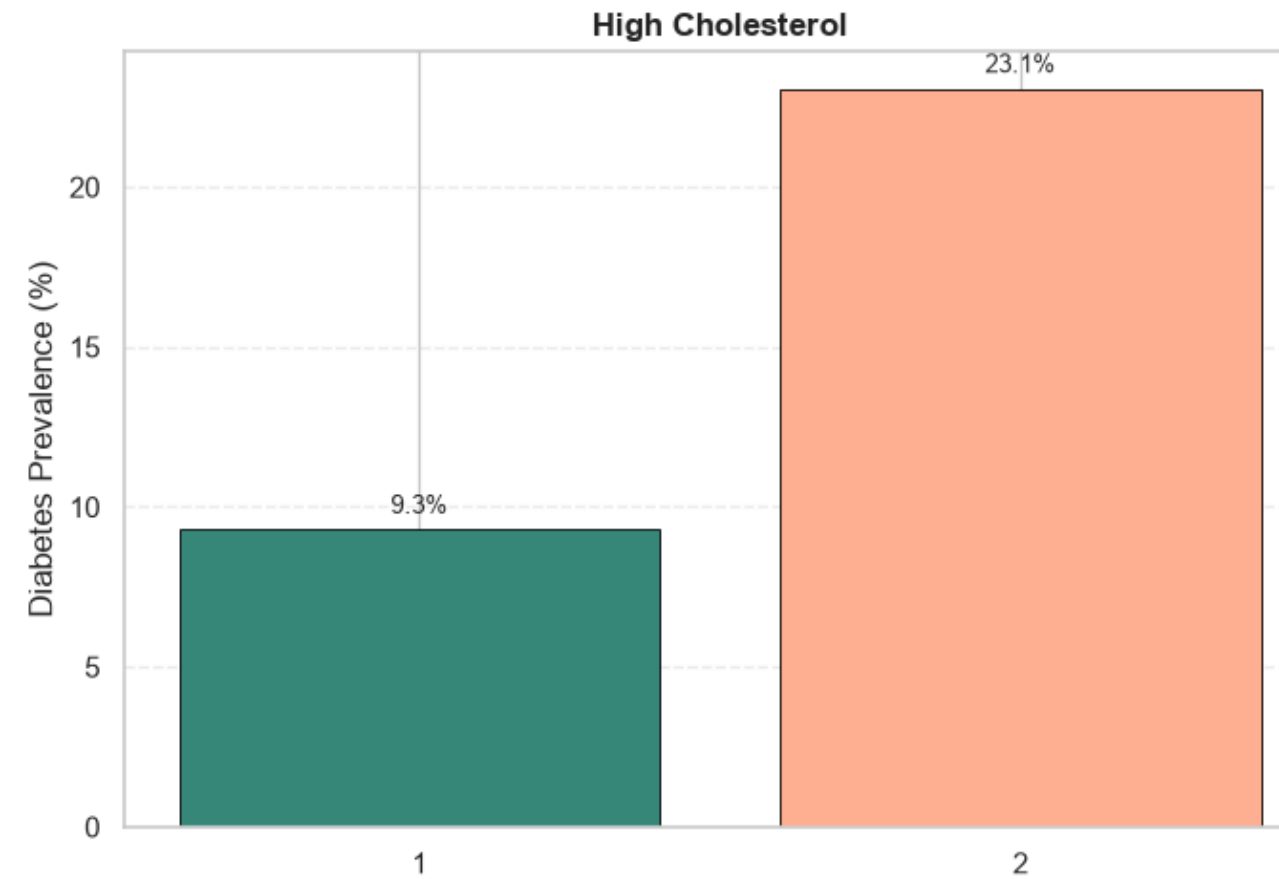
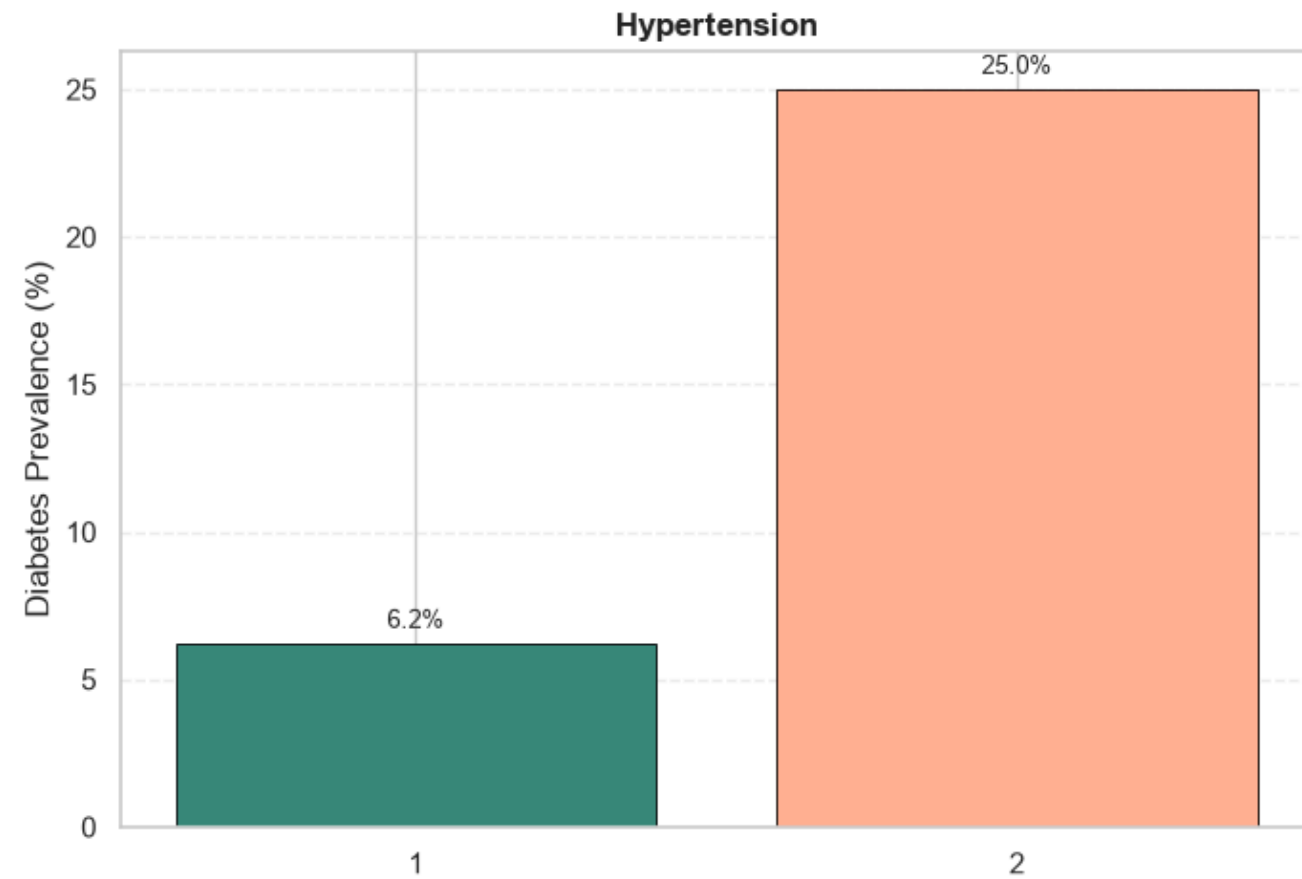
Target Variable Distribution: Type 2 Diabetes



- Class imbalance: 85.7% non-diabetic vs. 14.3% diabetic, consistent with real-world prevalence.
- Evaluation focus: **Recall** was prioritized over accuracy to reduce missed diabetes cases (false negatives).



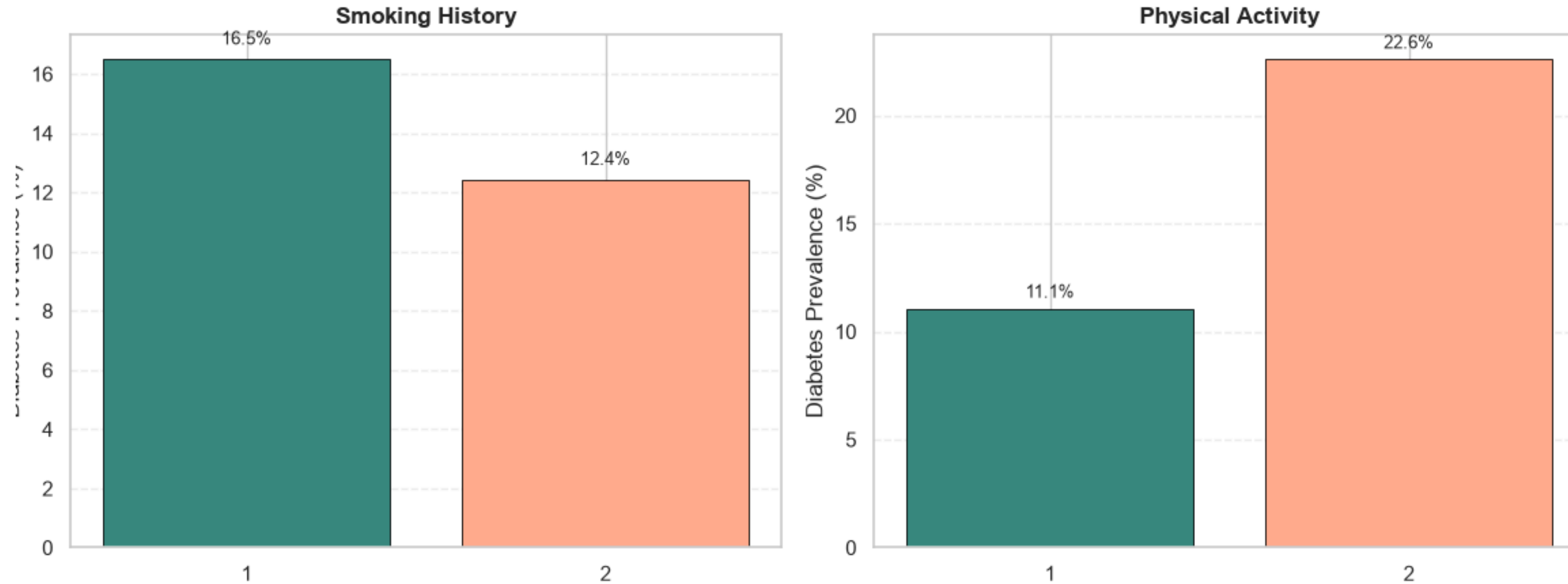
CLINICAL RISK FACTORS



- Diabetes prevalence was consistently higher among individuals with hypertension, high cholesterol, heart disease, and kidney disease.
- Kidney disease and heart disease showed the strongest association with diabetes.
- These findings align with clinical evidence, supporting their inclusion as key predictive features.

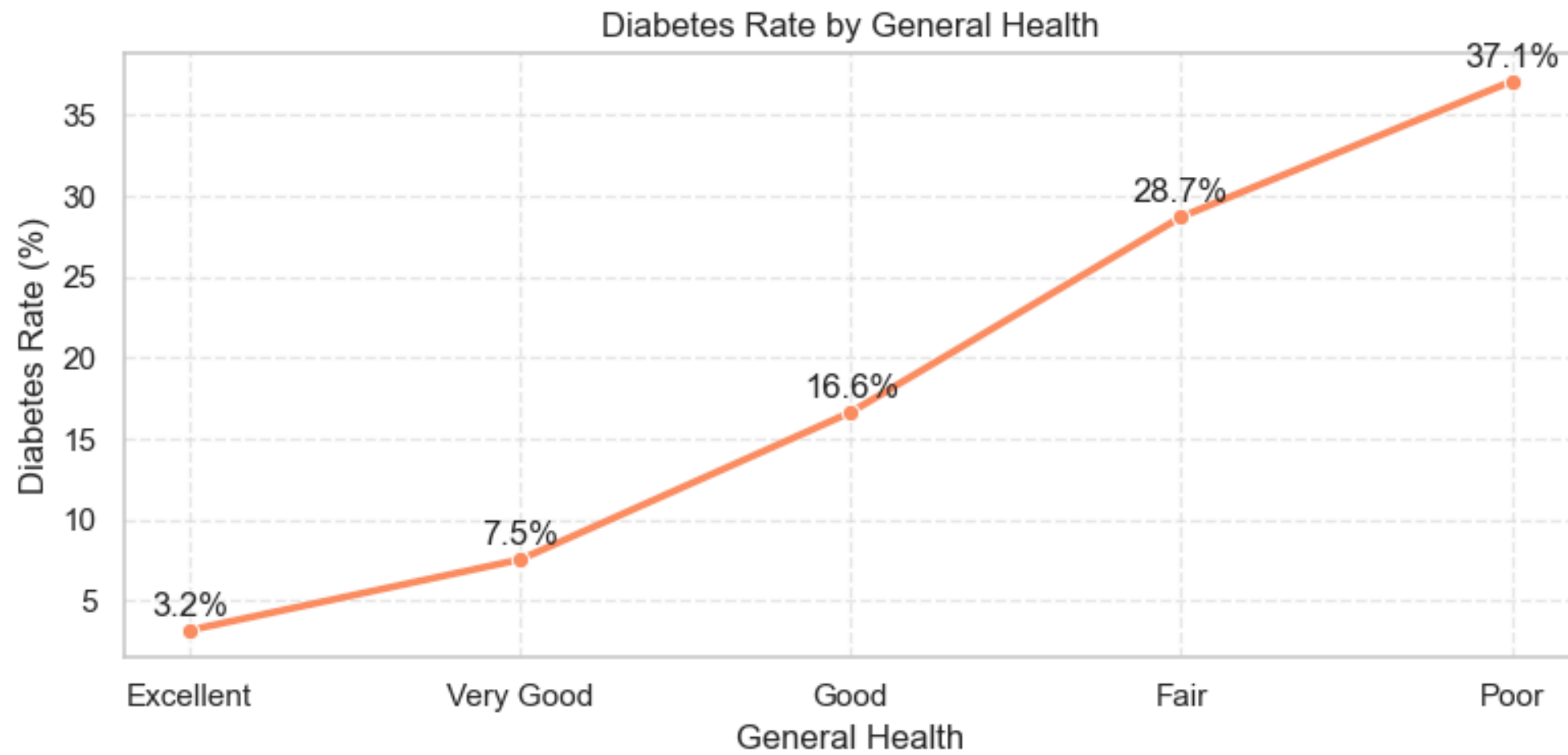
LIFESTYLE RISK FACTORS

Diabetes Prevalence by Lifestyle Risk Factors

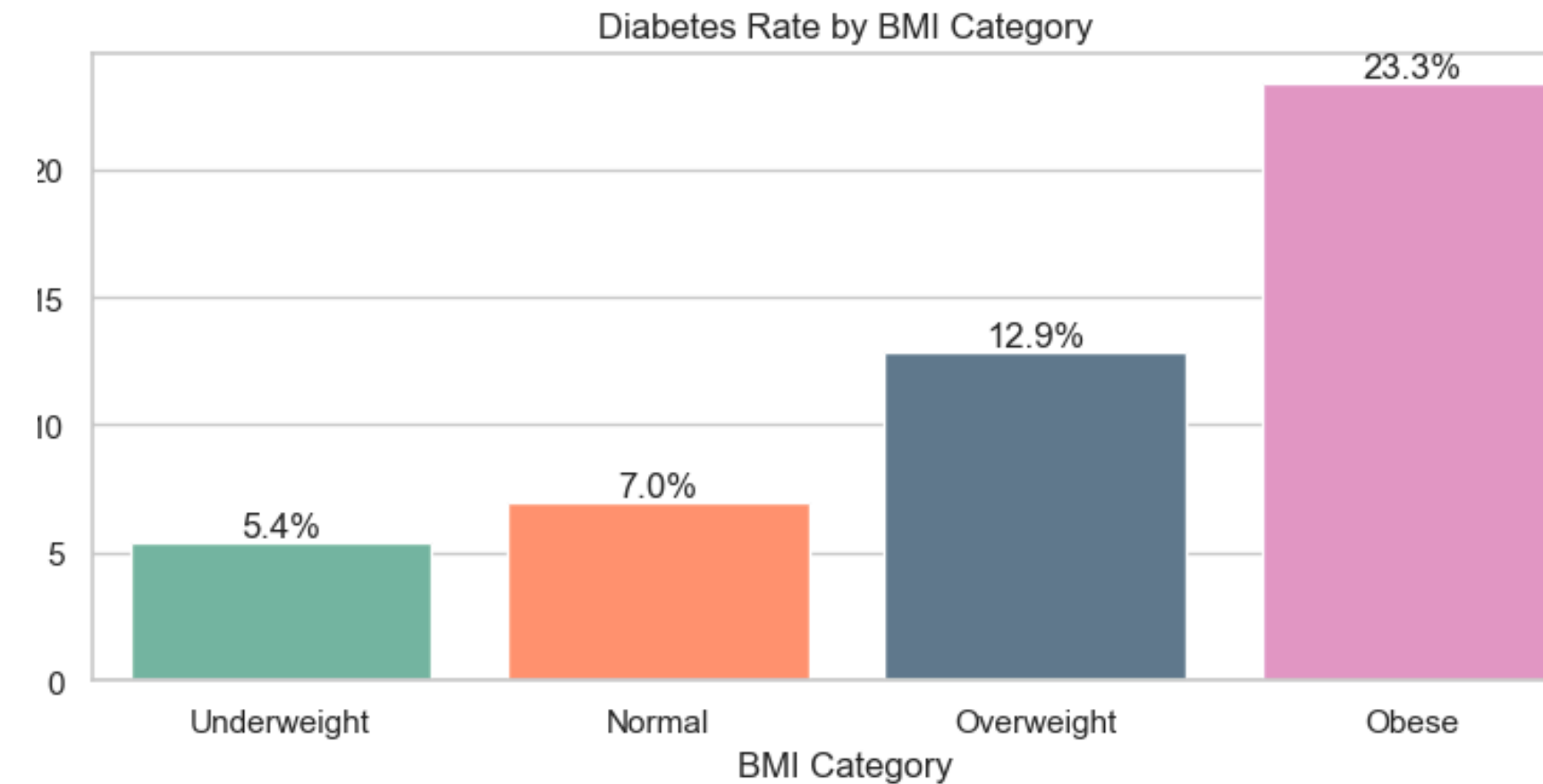


- Higher diabetes prevalence among physically inactive individuals.
- Smoking was associated with increased diabetes risk
- Supports lifestyle behaviors as important predictive features.

LIFESTYLE RISK FACTORS

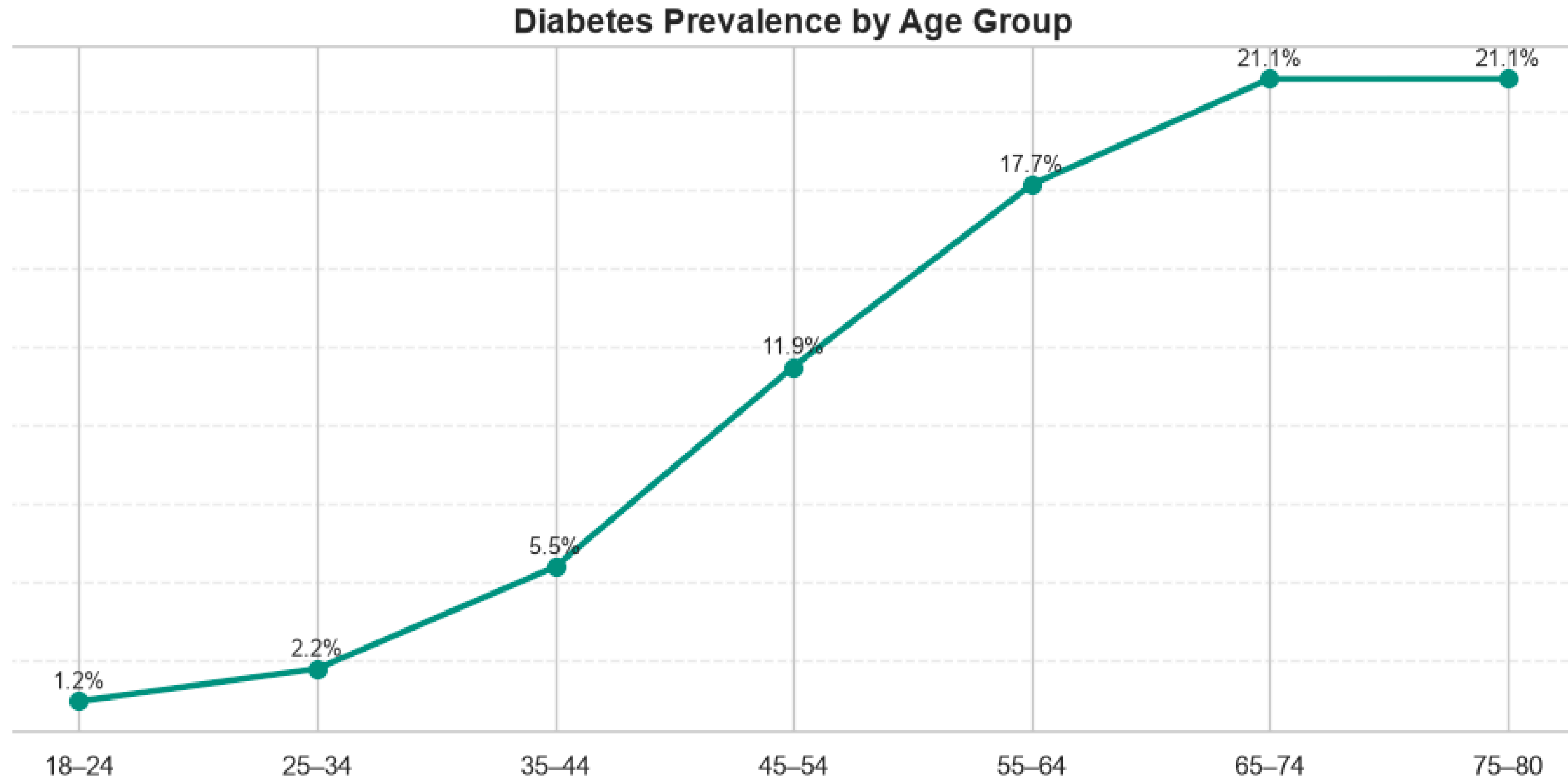


- The steady upward trend indicates a strong association between poorer general health and Type 2 Diabetes.



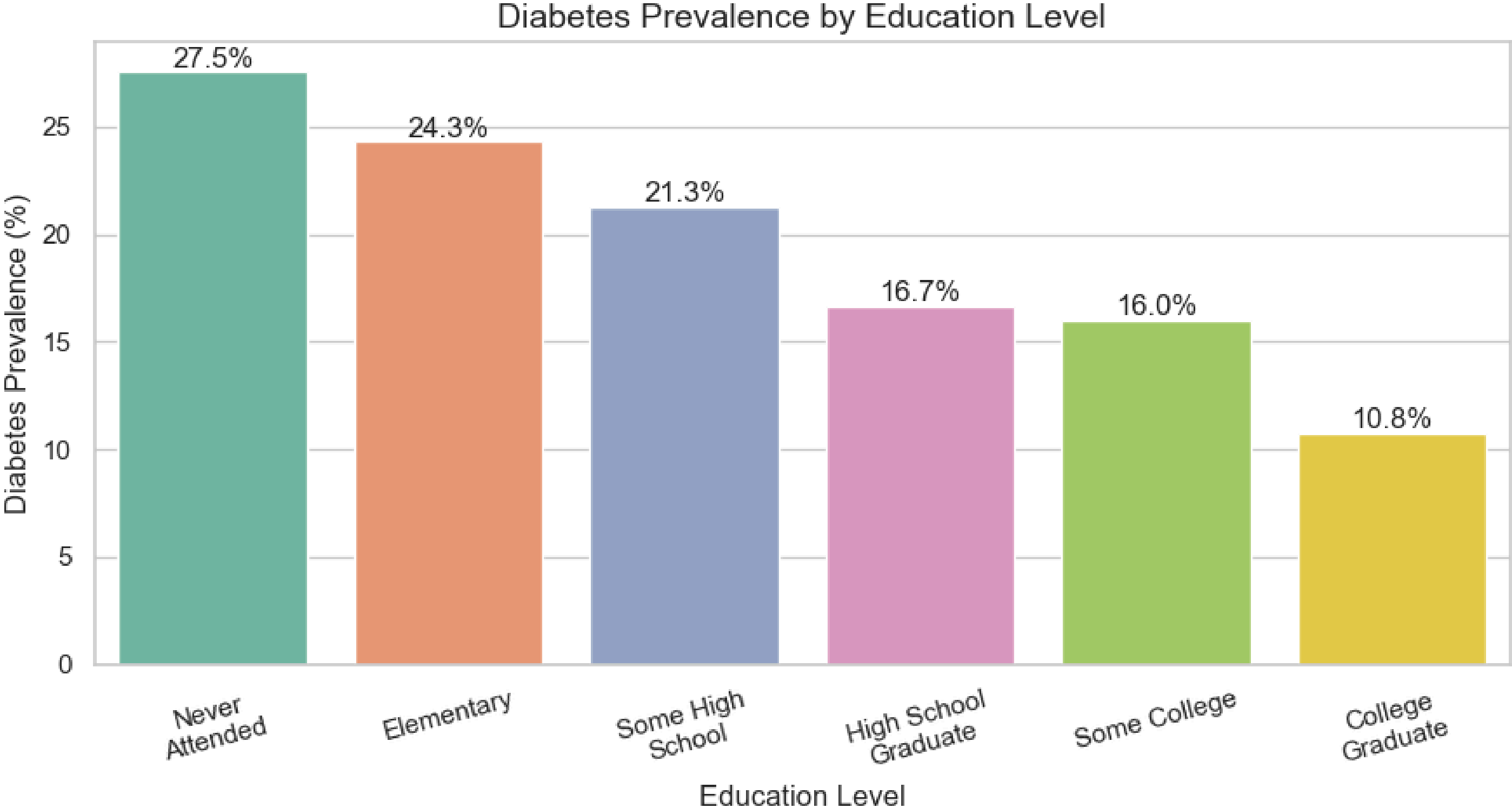
- The chart shows a clear upward trend between BMI category and diabetes rate. Diabetes prevalence increases from 5.4% among underweight respondents to 23.3% among respondents with obesity.

DEMOGRAPHIC RISK FACTORS



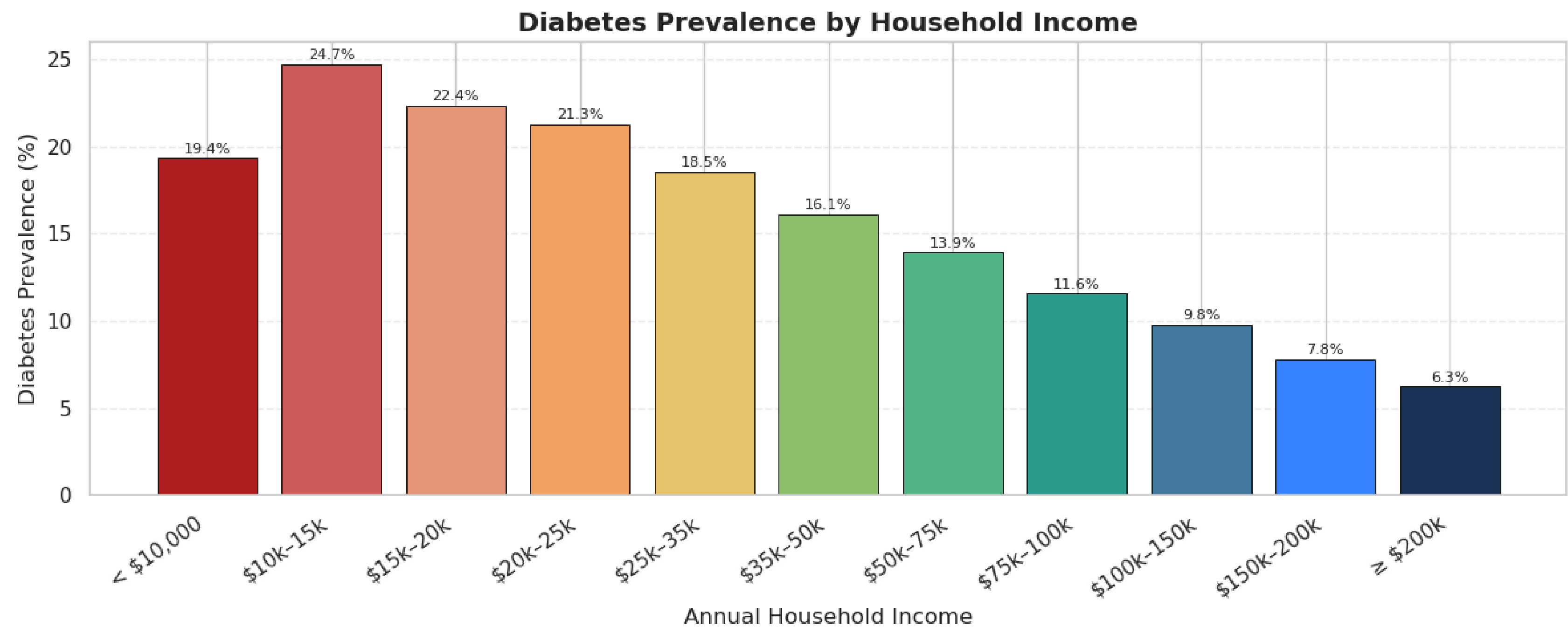
- The risk of Type 2 Diabetes generally increases with age due to reduced insulin sensitivity, progressive pancreatic β -cell dysfunction, and the accumulation of metabolic risk factors over time.

DEMOGRAPHIC RISK FACTORS



This pattern suggests that higher educational attainment may contribute to improved health literacy, healthier lifestyle choices, better employment opportunities, and greater access to healthcare services thus lowering the prevalence to Diabetes.

DEMOGRAPHIC RISK FACTORS



This pattern suggests that lower socioeconomic status is associated with a greater burden of Type 2 diabetes, potentially reflecting disparities in healthcare access, nutrition, education, and opportunities for maintaining healthy lifestyles.

CORRELATION OF FEATURES WITH DIABETES

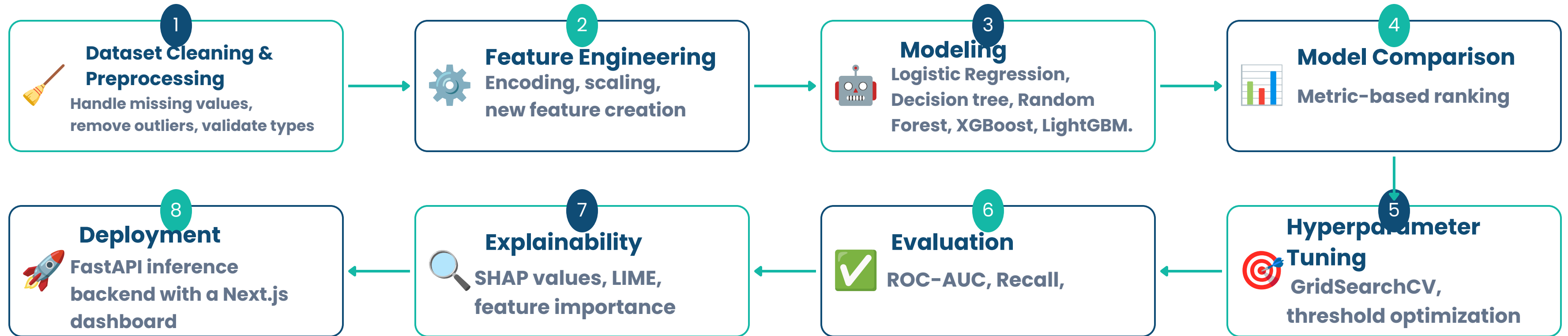


Key Observations

- Age, BMI, general health, and hypertension showed the strongest positive association with diabetes.
- Income and education exhibited weaker negative associations.
- The findings validate the selected predictors for machine learning model development.

Machine Learning Pipeline

End-to-End Workflow for Predicting Type II Diabetes Risk



Model Comparison Results

Success Metrics: **Recall** & **ROC-AUC** | Higher = Better at catching diabetic patients

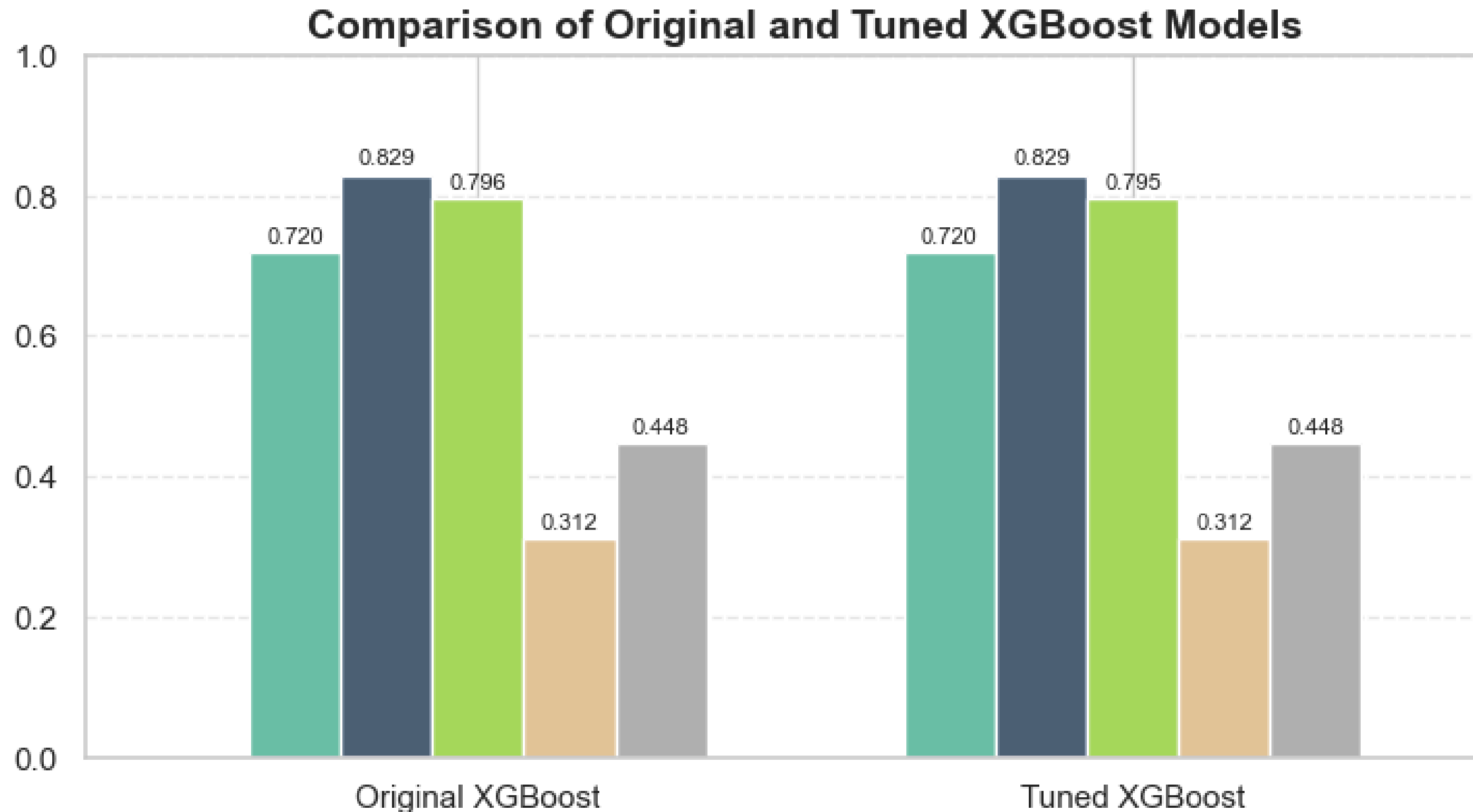
Model	Accuracy	ROC-AUC ★	Recall ★	Precision	F1-Score
XGBoost	0.7198	0.8291	0.7961	0.3117	0.4480
LightGBM	0.7157	0.8291	0.7991	0.3087	0.4454
Random Forest	0.7193	0.8239	0.7821	0.3092	0.4432
Logistic Regression	0.7291	0.8222	0.7639	0.3151	0.4462
Decision Tree	0.6803	0.8024	0.8057	0.2828	0.4187



XGBoost Selected — Best ROC-AUC (0.829) with strong Recall (0.796)

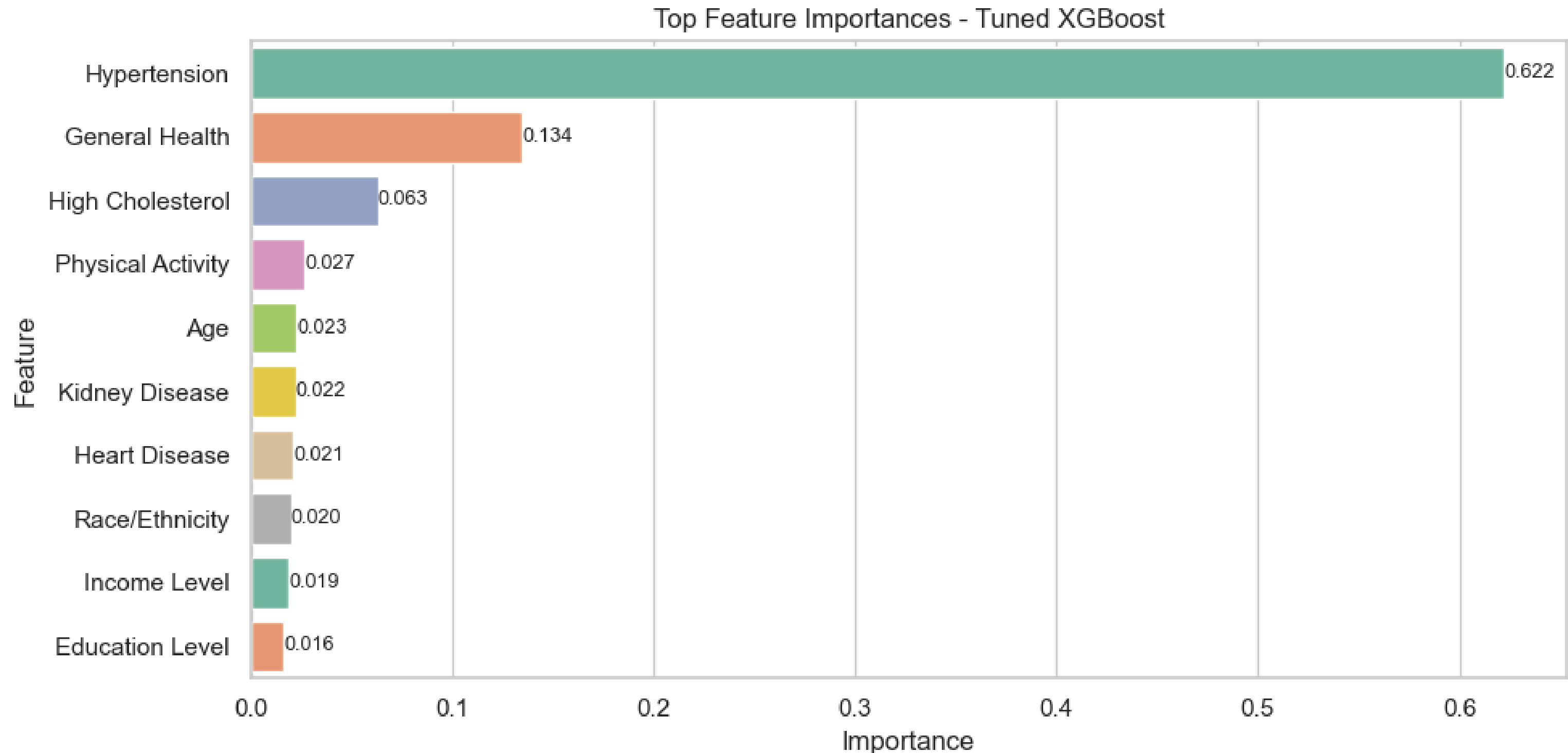
Decision Tree has highest Recall (0.806) but lowest ROC-AUC — XGBoost offers the best balance

Final Model Performance: XGBoost



The original XGBoost model was already well-optimized, and the tuned model did not provide a meaningful performance improvement.

Feature Importance



Top predictors: Hypertension, General Health, High Cholesterol

Additional predictors: BMI, Age, Kidney Disease, Heart Disease, Physical Inactivit

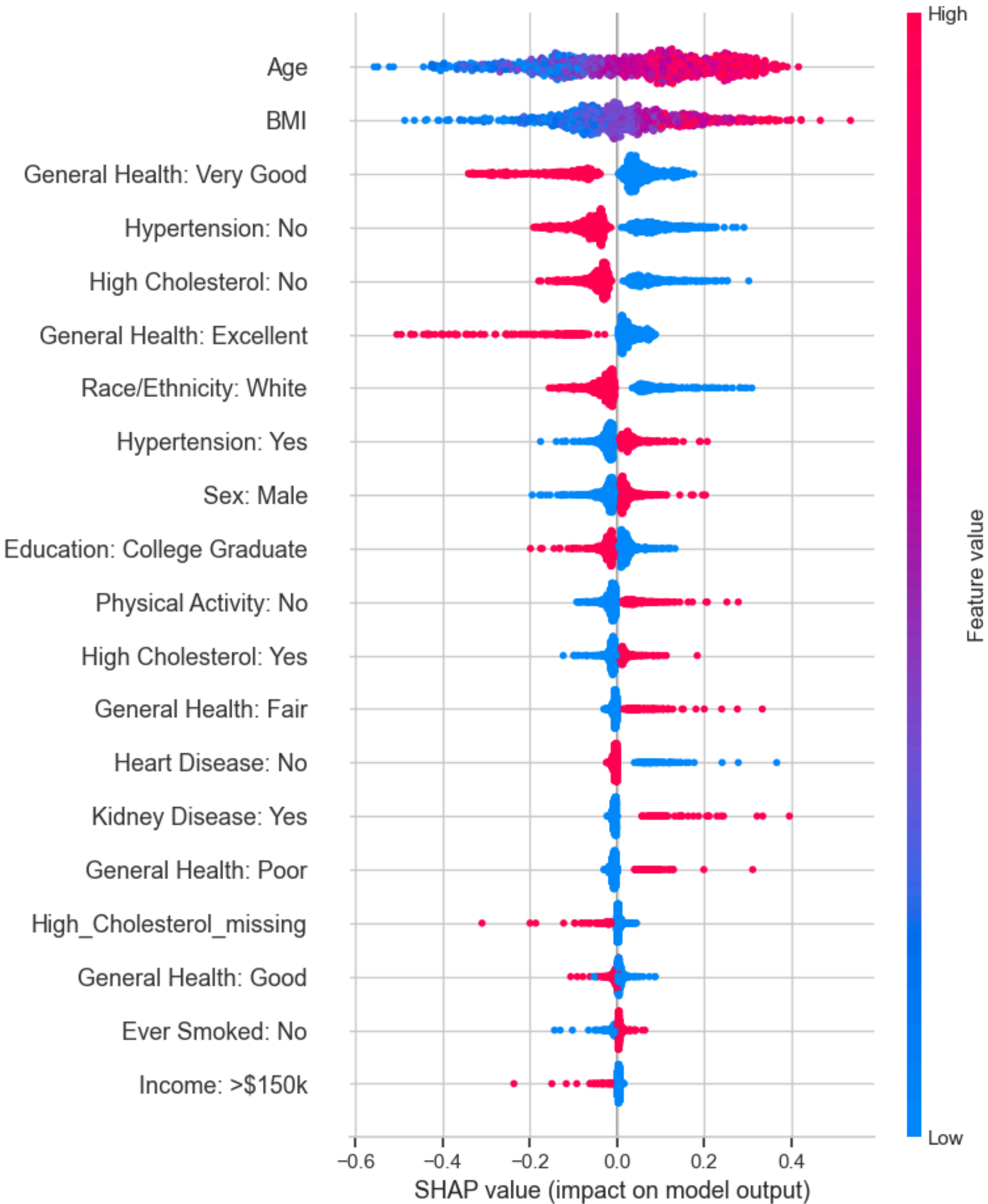
Cardiometabolic factors (cholesterol, BMI, kidney disease, heart disease, age) dominated feature importance.

Model Explainability: SHAP

Top 10 SHAP Features

Feature importance & direction of risk

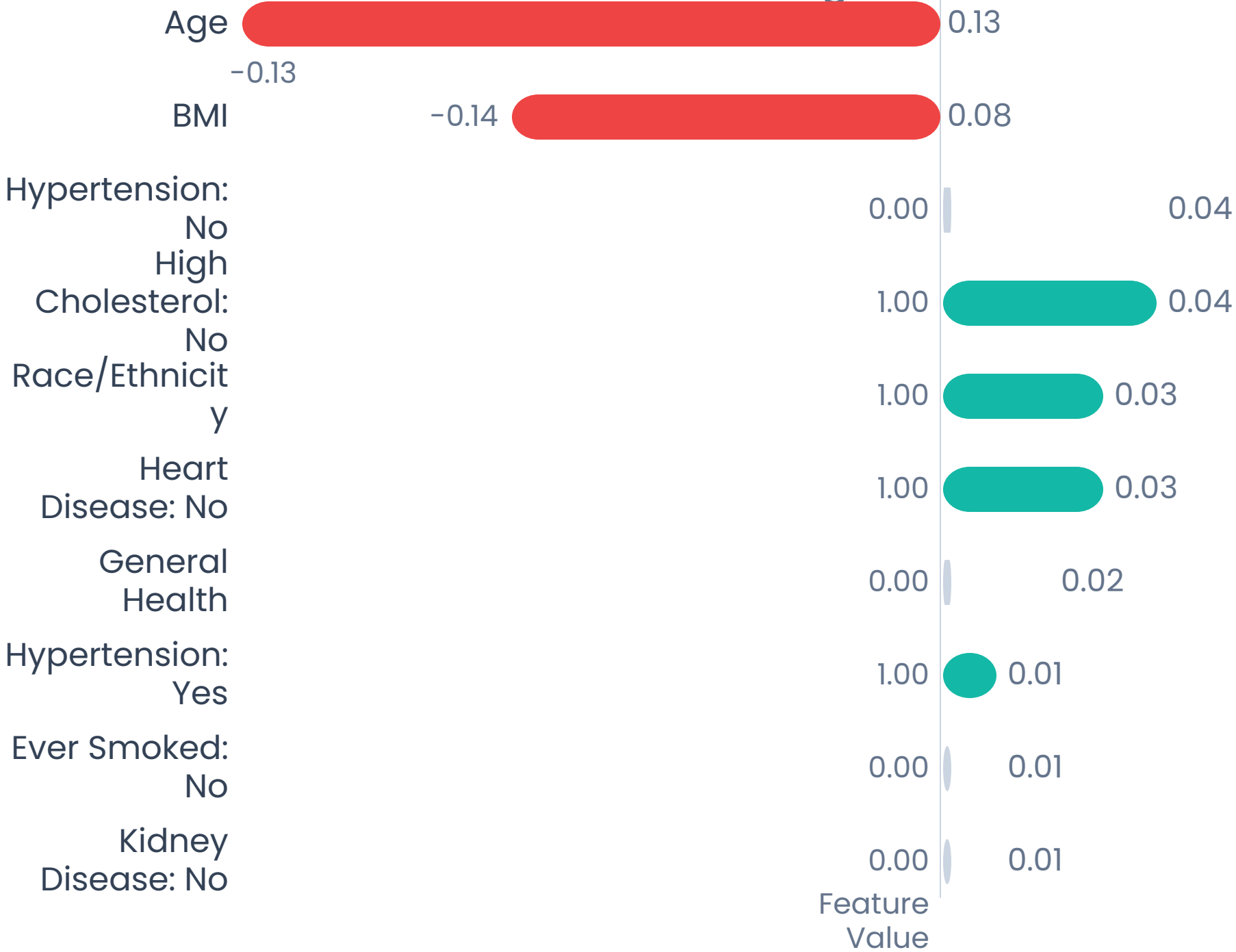
Feature	Importance	SHAP Insight
Age	Highest	↑ Older age increases risk
Hypertension	High	↑ Hypertension increases risk
BMI	High	↑ Higher BMI increases risk
General Health	High	↑ Poor health increases risk
High Cholesterol	High	↑ High cholesterol increases risk
Physical Activity	Moderate	↓ Activity reduces risk
Education Level	Moderate	↓ Higher edu slightly reduces risk
Income Level	Moderate	↓ Higher income slightly reduces risk
Heart Disease	Moderate	↑ Heart disease increases risk
Kidney Disease	Moderate	↑ Kidney disease increases risk



LIME Explanation — Individual Prediction

LIME Feature Contributions

Positive = increases diabetes risk | Negative = decreases risk



LIME Individual Prediction

68%

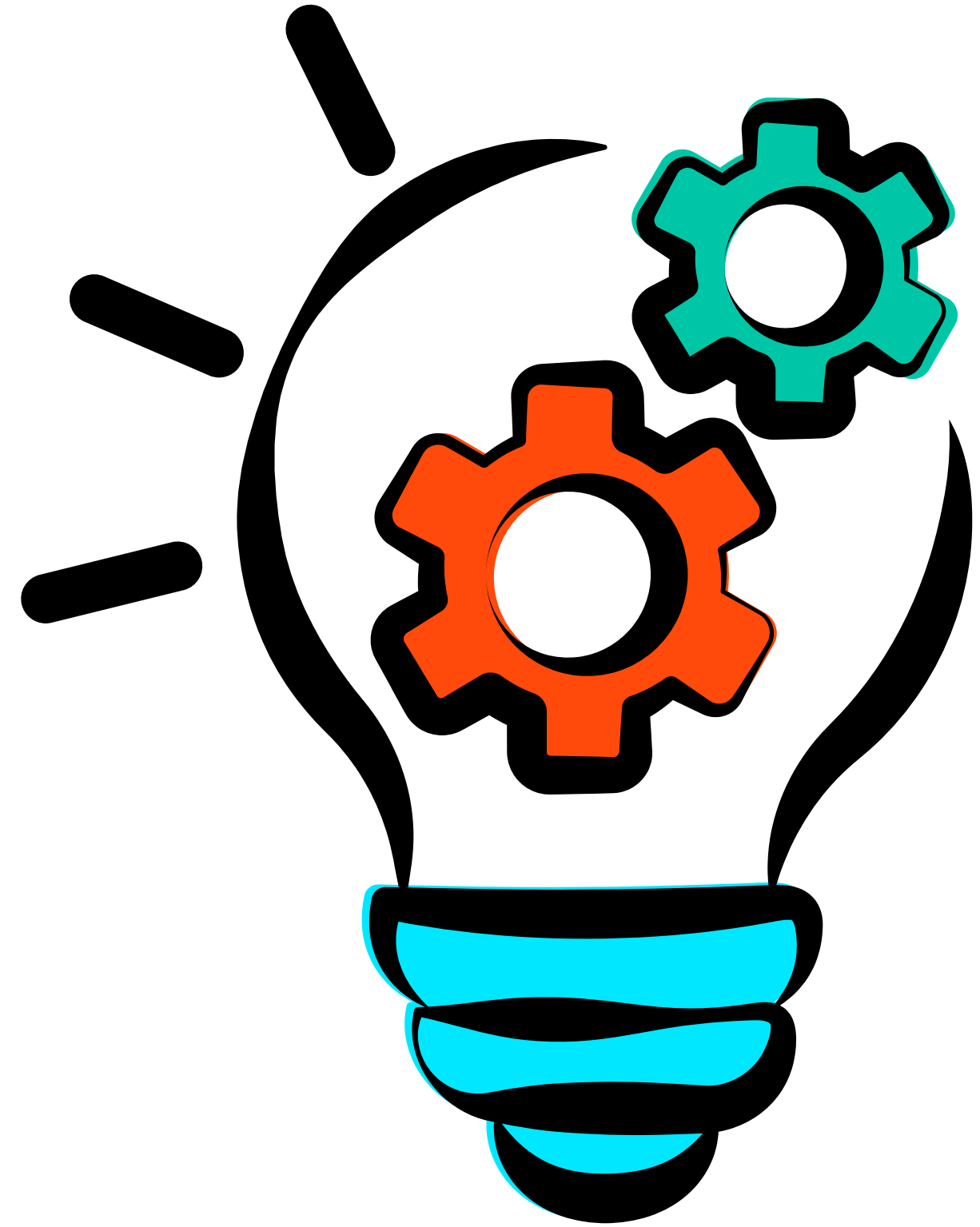
Predicted Diabetes Probability

Key Drivers

- Age and BMI are the strongest contributors to the prediction.
- Hypertension, cholesterol, race/ethnicity, and heart disease follow.
- General health, smoking, and kidney disease have smaller effects.

Key findings from the diabetes prediction study

- Age was the strongest predictor of Type 2 Diabetes, with risk increasing among older individuals.
- Hypertension, high BMI, poor general health, and high cholesterol were the key clinical risk factors.
- Physical activity reduced diabetes risk, while higher education and income were associated with slightly lower risk.
- SHAP identified age, hypertension, BMI, general health, and high cholesterol as the most influential features.
- The findings demonstrate that ML can support early identification of individuals at high risk of Type 2 Diabetes.



Limitations & Challenges



Self-reported BRFSS data may contain recall bias and measurement errors



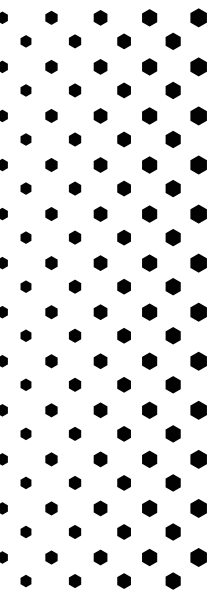
Model generalizability limited to US population; external validation needed

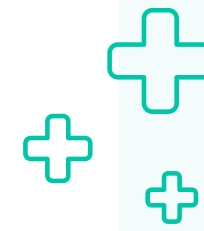


Class imbalance and missing data affect prediction reliability



EHR integration and clinical workflow adoption pose deployment barriers





RECOMMENDATIONS

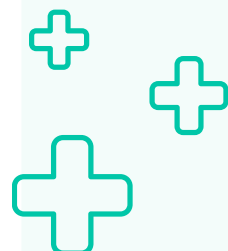
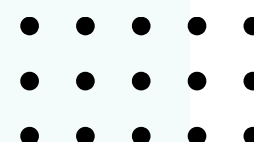
+ Prioritize screening for adults with multiple clinical risk factors

+ Use the model as a risk-stratification support tool, not a diagnosis.

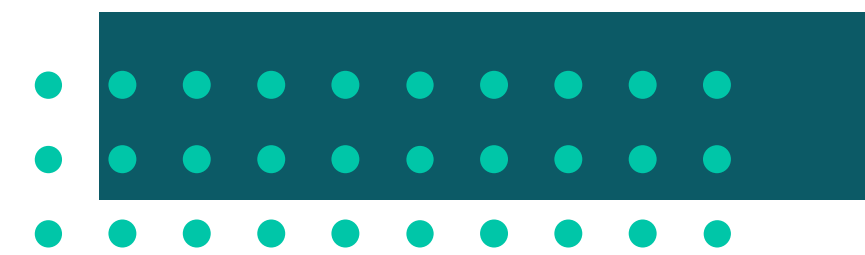
+ Tune the decision threshold to the operational goal.

+ Pair model outputs with intervention pathways.

+ Monitor fairness and subgroup performance.



Conclusion



79.61% Recall

82.93% ROC-AUC

~8 in 10 Detected



Predictive Power

Type 2 Diabetes risk predicted from 14 routine health survey variables without requiring any laboratory tests.



Explainability

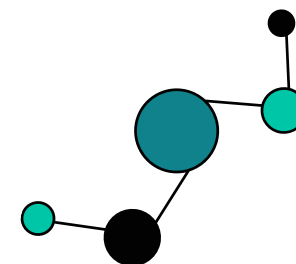
SHAP and LIME transform each prediction into a patient-specific risk profile clinicians can interpret and communicate.



Deployment

FastAPI + Next.js platform with single/batch prediction, analytics, and PDF reports — ML as a practical clinical tool.

Machine learning, applied responsibly and transparently, has significant potential to improve early detection of chronic diseases and reduce the burden of preventable illness at population level.





Any Question?

Thank You!

